

Numerical and Experimental Study on the Motion Response of a Sea-Launch Platform Under High Impact Force

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Abstract

The impact force effect on launch platform motion response represents a critical safety consideration that requires thorough investigation prior to sea-launch implementation. This paper examines a self-designed semi-submersible launch platform comprising a box-shaped deck, six columns, and two pontoons, with sufficient structural stiffness to be analyzed as a rigid body. A proprietary code based on three-dimensional linear potential theory was developed for hydrodynamic analysis of the launching process. The Cummins equation was implemented to calculate platform responses under substantial impact force. The numerical results were validated through comparison with ANSYS/Aqwa commercial software for platform motion response in both launch and non-launch cases. Additionally, two model tests were conducted in a sea-keeping wave basin at a scale ratio of $\lambda = 1:40$. The numerical results demonstrated good agreement with experimental data. Both numerical and experimental findings indicate that platform motion responses result from wave-induced effects and impact force/rocket weight effects, with the latter typically predominant. Numerical simulations revealed that in moderate sea states, maximum heave and pitch motions measure 0.6 m and 1° , respectively, suggesting the viability of sea-launch operations using the designed platform under these conditions.

Key words: sea-launch platform, impact force, wave basin experiments, numerical simulations

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1 Introduction

Sea-based launch offers several advantages over land-based alternatives, including mobility and flexibility in launch site selection, simplified rocket transportation, and reduced risk of surrounding area damage from rocket debris, thereby garnering increasing industry attention (Song et al., 2021). Sea-based launches encompass two categories: cold launch, requiring secondary ignition after deck ejection, and thermal launch, utilizing single ignition with benefits of simplified launch systems and enhanced payload capacity per unit space (Konyukhov, 2004). However, thermal launch implementation presents challenges due to significant rocket wake impact force on the platform, potentially causing equilibrium position deviations and substantial translational and angular displacements (Wang et al., 2022). These deviations may adversely affect spacecraft attitude control during takeoff and significantly influence platform motion

response (Mu et al., 2023). Therefore, quantitative analysis of platform motion under impact force is essential for determining appropriate operational conditions to ensure launch success and platform safety.

Potential theory and its derivative numerical methods are commonly employed for simulating and predicting offshore platform hydrodynamic response. Zhang et al. (2020) investigated second-order hydrodynamic effects on floating wind turbine platforms, comparing frequency and time domain analyses across three platform configurations. Pham and Shin (2020) integrated potential method with NREL FAST for fully coupled wind and wave simulations, achieving results consistent with model experiments. Zou et al. (2022, 2024) and Chen et al. (2024) employed an efficient constant parameter time-domain model (CPTDM) to examine multi-body system nonlinear dynamics, demonstrating superior accuracy and efficiency particularly at resonant frequency

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compared to alternative time-domain simulations. However, complex phenomena such as green water and slamming in severe sea states necessitate model experiments or CFD simulations. Rivera-Arreba et al. (2019) utilized second-order potential-flow based solver and fully nonlinear Navier-Stokes equations solver to analyze semi-submersible floating platform response in non-steep waves, revealing potential flow model limitations in accurately capturing heave response and viscous effect relationships at natural frequency excitation. Paulsen et al. (2019) conducted physical model tests examining wave slamming effects on monopile wind turbine foundations, establishing probability formulas for wave slamming and associated force. Zhu et al. (2022) and Huo et al. (2023) performed model tests and CFD simulations investigating floating wind turbine slamming characteristics under freak and irregular waves respectively.

Unlike floating wind turbines or offshore drilling platforms, offshore launch platforms are subject to substantial instantaneous impact loads during rocket launches. Understanding the platform's motion response under these impact loads is essential for the overall layout design and critical for ensuring safe and stable operations. When a rocket launches from an offshore platform, the platform deviates from its equilibrium position and experiences significant translational and angular displacements. These displacements affect both spacecraft attitude control during takeoff (Cui and Hu, 2007) and the platform's motion response. Consequently, quantitative analysis of the platform's motion behavior under impact forces is necessary to determine suitable operational conditions for rocket launches and ensure platform safety. Several studies have examined the hydrodynamic response characteristics of floating platforms under impact loads from spacecraft launches. Wang et al. (2003) investigated ship motion characteristics in waves under impact force through model tests, validating their established Kane's dynamic model (Wang et al., 2001). Yu et al. (2021) analyzed the hydrodynamic response of offshore launch platforms under various conditions using ANSYS/Aqwa three-dimensional potential commercial software, examining how factors such as draft, impact load, and launch phase influence platform motion response. Pang et al. (2023) developed a test method for evaluating impact vibration

acceleration response of offshore rocket launch platforms, studying launch platform impact response during cold launch through finite element analysis and shipboard experiments. Their findings revealed rapid acceleration peak decay during ejection, followed by oscillating acceleration signals during ignition. Hou (2024) employed STAR-CCM+ to examine motion responses of various offshore launch platform types under different wave conditions, launch positions, and platform motion phases.

Currently, limited research exists regarding spacecraft launch platform motion responses under impact loads in complex marine environments. Engineering practice requires a method for rapid prediction of launching platform hydrodynamic responses to impact loads in waves.

This paper introduces a self-designed offshore launch platform and validates its launch mission feasibility through numerical and experimental methods. Furthermore, an in-house code based on linear three-dimensional potential theory is developed to calculate the platform's hydrodynamic response, accounting for both wave excitation and launch impact force effects. The validity and accuracy of the in-house code are verified through comparisons with ANSYS/Aqwa commercial software and model test results, demonstrating the method's applicability for relevant engineering applications.

This paper is arranged as follows. In Section 2, the designed launch platform is presented. In Section 3, the model test and numerical method are presented respectively, and the in-house code is verified. In Section 4, the hydrodynamic calculations of the launch platform under different sea and working conditions are carried out, and the hydrodynamic characteristics of the launch platform are analyzed in conjunction with the results of the model tests. Finally, conclusions and prospects for future work are given in Section 5.

2 Problem description

The launch platform examined in this study is a self-designed semi-submersible platform comprising a box-shaped deck and six columns supported by two pontoons, as shown in Fig. 1. During normal operations, the pontoons remain submerged with only the columns piercing the water surface. Table 1 presents the principal parameters of the self-

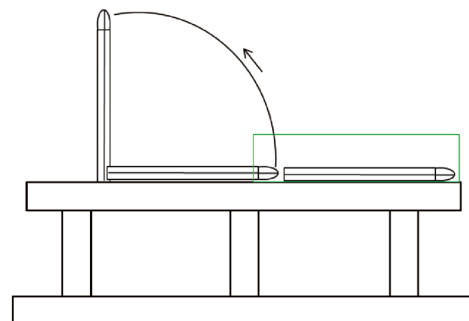


Fig. 1. Sketch of semi-submersible launch platform.

Table 1 Principal parameters of the launch platform under operating condition

Parameter	Prototype value
Length (m)	151.2
Breadth (m)	70
Depth (m)	38
Draft (m)	21
Longitudinal location of the gravity center (from the stern) (m)	73.877
Vertical location of the gravity center (above the bottom base of the platform) (m)	16.27
Transverse moment of inertia I_{xx} ($\text{kg} \cdot \text{m}^2$)	5.14×10^{10}
Longitudinal moment of inertia I_{yy} ($\text{kg} \cdot \text{m}^2$)	1.19×10^{11}
Yaw moment of inertia I_{zz} ($\text{kg} \cdot \text{m}^2$)	1.42×10^{11}

designed platform.

Under non-operating conditions, the rocket is stored horizontally in the storeroom, while under operating condition it is positioned vertically on the deck near the platform's stern. This investigation employs both numerical simulation methods and wave basin model tests, focusing on platform responses to waves and impact forces during three operational stages: pre-launch, during launch, and immediately after launch.

3 Experimental and numerical modeling

3.1 Model test

3.1.1 Facilities and equipment

The semi-submersible sea launch platform underwent testing in the seakeeping wave basin at China Ship Scientific

Research Center (CSSRC), as shown in Fig. 2. The basin measures 69 m in length, 46 m in width, and 4 m in depth. The wave generation system comprises 188 segments of flap-type wave makers installed along the adjacent short and long sides of the basin, corresponding to the right and bottom sides in the right of Fig. 2. For wave absorption, cement gratings on a 7 m long parabolic arc are installed along the sides opposite to the wave generators. The wave absorber is positioned on the left and upper sides as shown in the right of Fig. 2. The wave basin can generate regular waves with a maximum height of 0.5 m and irregular waves up to 0.4 m. The facility is equipped with a 6-DOF non-contact motion measurement system and servo-type wave height meters to measure vessel motion and wave height.

3.1.2 Experiment set-up

Based on the wave basin dimensions and wave generator capacity, a scale ratio of $\lambda=1:40$ was selected for the platform model depicted in Fig. 3. The model's main structure was constructed from fiber reinforced plastics (FRP) and maintained the identical configuration shown in Fig. 1. During testing, the platform was positioned at the basin's center, with a dynamic positioning system restricting surge, sway, and yaw motions within acceptable limits (Sun and Yin, 2023; Hu et al., 2022). The platform's superstructure geometry closely resembled that of the prototype. Prior to testing, the model's weight, center of gravity, and mass moments of inertia were measured and calibrated to correspond with prototype specifications. The experiment maintained identical Froude and Strouhal numbers between the model and

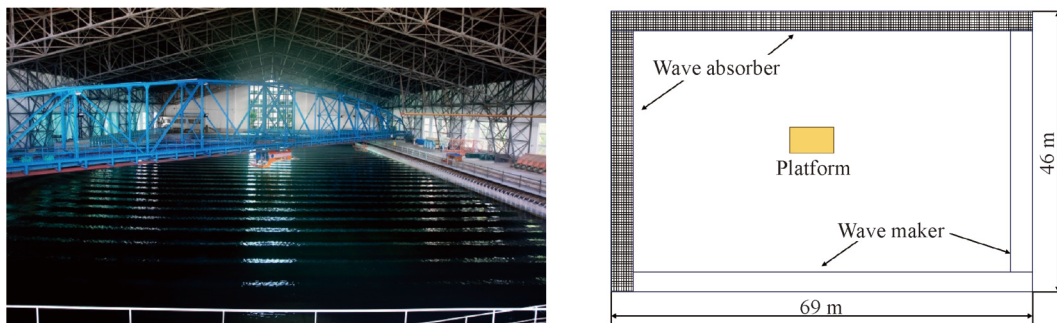


Fig. 2. Seakeeping wave basin in CSSRC (CSSRC, 2021; ITTC, 2017).

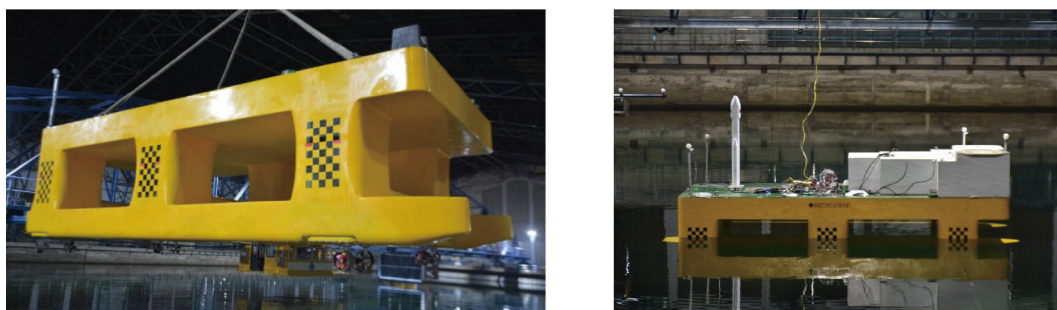


Fig. 3. Launch platform model in experiment.

prototype, ensuring gravitational and inertial force similarities. The rocket was positioned vertically on the platform deck before launch and subsequently propelled skyward using a small solid rocket motor, while the platform's motion response was recorded throughout the entire process.

3.2 Numerical method and validation

3.2.1 Numerical method

This study employs time domain dynamic simulations based on linear three-dimensional potential theory to investigate platform motions under substantial impact forces. Both commercial software and a self-developed code were utilized to perform numerical simulations and compare with model test results. The global earth fixed coordinate system ($O_0-X_0Y_0Z_0$) originates at the mean free surface, while the local body fixed coordinate system ($O-XYZ$) originates at the projection of the center of gravity on the mean free surface, as defined in Fig. 4. The wave direction is designated as 0° when traveling along the X -axis and 90° when traveling along the Y -axis. The wetted surface edges of the numerical model differ slightly from the experimental model for simplification purposes, which is not expected to significantly affect platform responses. All simulations maintain zero forward speed. The numerical simulations utilize the prototype launch platform parameters.

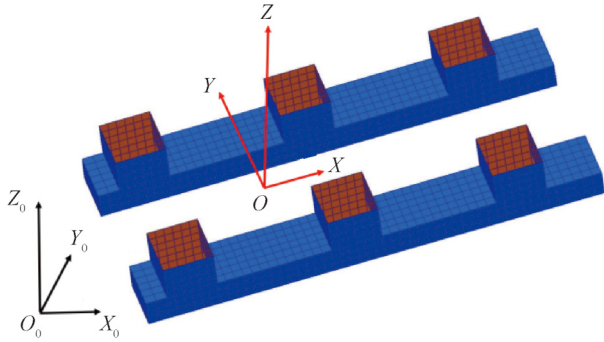


Fig. 4. Wet surface of launch platform.

The equations of motion were established in the self-developed in-house code for the following three cases:

(1) Regular wave without impact force

The equation of motion in the time domain can be derived directly by converting the equation in the frequency domain for regular waves:

$$[m + A(\omega)] \ddot{X}(t) + [b + B(\omega)] \dot{X}(t) + KX(t) = F_h(t) + F_d(t) + F_g(t), \quad (1)$$

where m represents the structural mass matrix, b denotes the artificial linear damping matrix, K indicates the hydrostatic stiffness matrix, $F_h(t)$ represents the hydrodynamic wave incident force, $F_d(t)$ denotes the wave diffraction force, and $F_g(t)$ represents the gravitational force. $A(\omega)$ and

$B(\omega)$ represent the frequency-dependent added mass and damping coefficient, respectively. ω represents the angular frequency of the wave, and t denotes time.

(2) Irregular wave without impact force

The motion of the in-house code in irregular waves follows Cummins equation (Pang et al., 2023) and which is expressed in a convolution integral form:

$$(m + A_\infty) \ddot{X}(t) + b \dot{X}(t) + KX(t) + \int_0^t h(t-\tau) \ddot{X}(\tau) d\tau = F_h(t) + F_d(t) + F_g(t), \quad (2)$$

where A_∞ represents the added mass matrix at infinite frequency and h denotes the acceleration impulse-response function matrix defined by

$$h(t) = \frac{2}{\pi} \int_0^\infty B(\omega) \frac{\sin(\omega t)}{\omega} d\omega = \frac{2}{\pi} \int_0^\infty [A(\omega) - A_\infty] \cos(\omega t) d\omega. \quad (3)$$

From the above equations, after calculating $A(\omega)$, both $h(t)$ and A_∞ can be determined.

(3) Regular and irregular waves with impact force

The equation of motion incorporating an impact force in both regular and irregular waves can be unified and expressed as:

$$(m + A_\infty) \ddot{X}(t) + b \dot{X}(t) + KX(t) + \int_0^t h(t-\tau) \ddot{X}(\tau) d\tau = F_h(t) + F_d(t) + F_g(t) + F_{\text{impact}}(t), \quad (4)$$

where F_{impact} represents the time-varying non-periodic impact force. It should be noted that most commercial software and open-source codes utilize Eq. (1) for regular wave cases. However, since the launch impact is a transient process and non-periodic in nature, the platform's response is inherently non-periodic; therefore, the convolution form becomes essential in calculating the radiation force, even for regular waves. Eq. (4) is solved as:

$$(m + A_\infty) \ddot{X}(t) = F_h(t) + F_d(t) + F_g(t) + F_{\text{impact}}(t) - \left(b \dot{X}(t) + KX(t) + \int_0^t h(t-\tau) \ddot{X}(\tau) d\tau \right) \quad (5)$$

with fourth-order Runge-Kutta time-stepping scheme.

The simulation of the launch process began with frequency domain hydrodynamic analysis to determine the platform's hydrodynamic parameters, including wave excitation forces, added mass, and radiation damping. These parameters were subsequently converted into time domain wave forces corresponding to the specified wave spectrum conditions. The software Thafits (Taihu laboratory of deep-sea technological science, 2025), developed by CSSRC, was utilized for the frequency domain hydrodynamic analysis. For comparative purposes, the commercial software ANSYS/Aqwa was also employed.

3.2.2 Validation of numerical model without impact force

The frequency domain results obtained from Thafts were compared with Aqwa outputs, and time domain simulations were conducted using the in-house code (as Thafts currently lacks time domain functionality) without impact force for comparison with Aqwa. The frequency domain analysis employed 2374 nodes and 2304 meshes, determined through a convergence test. Artificial diagonal linear damping was implemented with the values of 3.25×10^6 N/(m/s), 2.8×10^9 Nm/(rad/s), 1.44×10^{10} Nm/(rad/s) for the heave, roll and pitch motions, respectively, corresponding to 3%, 5% and 8% of the critical damping.

The Response Amplitude Operators (RAOs) of the launch platform were calculated for various wave directions using both Thafts and Aqwa, yielding consistent results. Fig. 5 presents the RAOs at infinite water depth and 0° wave direction, where black lines indicate Thafts results and red circles represent Aqwa results. The data demonstrate exact correspondence between the two software across the common wave frequency range.

The platform motion time history was obtained using the time domain in-house code. The results were compared with Aqwa calculations for various wave directions. Fig. 6 presents the results for regular waves with constrained sway, surge, and yaw motions. The heave and pitch motions at 0° and 90° wave directions are shown for frequencies ranging from 0.2 to 1.6 rad/s. The solid lines indicate Aqwa

results, while the dashed lines represent the in-house code calculations. The comparison in Fig. 6 demonstrates that the hydrodynamic coefficients and diffraction force are accurately transformed from the frequency domain to time domain in the in-house code.

The decay simulation in still water was conducted using both the in-house code and Aqwa based on Eq. (2), with the hydrodynamic force set to zero. The kernel function of the convolution appears in Fig. 7a. During the simulation, all motions except heave were constrained, and the results are displayed in Fig. 7b. The numerical results from Aqwa and the in-house code demonstrate identical outcomes.

To validate the accuracy of the in-house code for irregular wave cases, a JONSWAP spectrum was implemented in the numerical simulation of the platform responses. The wave spectrum is expressed as:

$$S(\omega) = \frac{5}{16} A H_s^2 \omega_p^4 \omega^{-5} \exp \left[-\frac{5}{4} \left(\frac{\omega}{\omega_p} \right)^{-4} \right] \gamma \exp \left[-0.5 \left(\frac{\omega - \omega_p}{\sigma \omega_p} \right)^2 \right], \quad (6)$$

where ω_p represents the peak frequency in rad/s, γ denotes the peak enhancement factor, H_s indicates the significant wave height, σ is defined as:

$$\sigma = \begin{cases} 0.07, & \omega \leq \omega_p \\ 0.09, & \omega > \omega_p \end{cases} \quad (7)$$

and

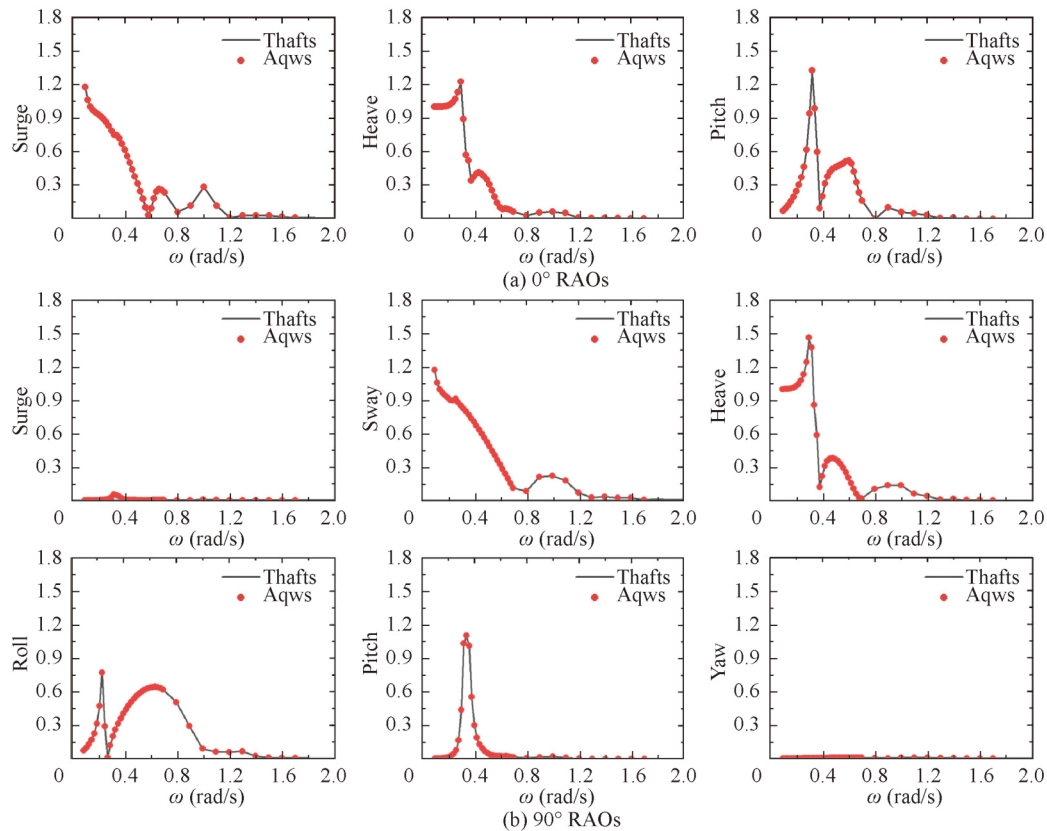


Fig. 5. RAOs of the launch platform.

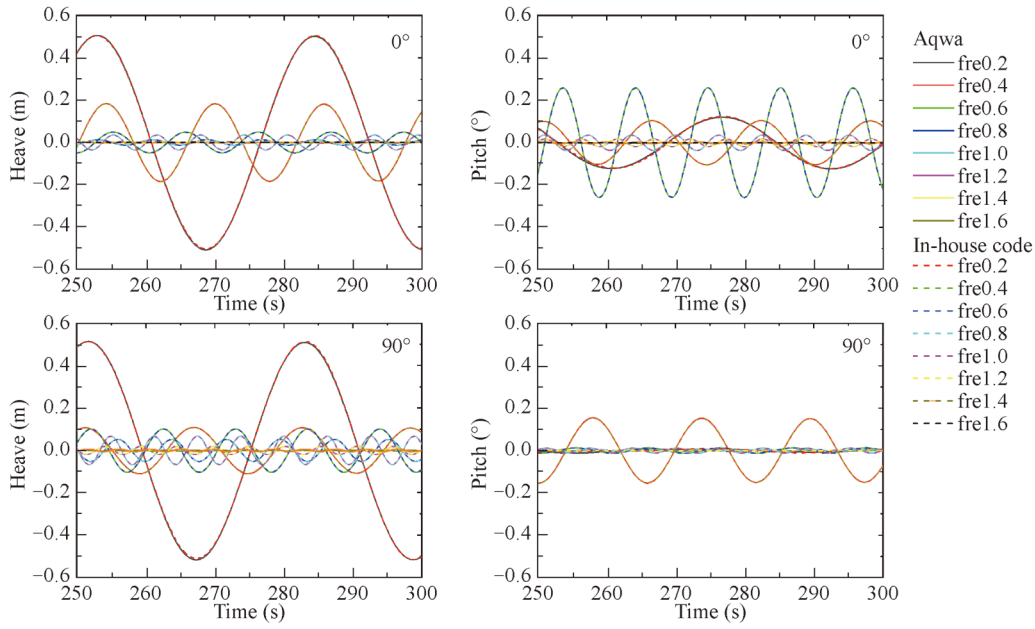


Fig. 6. Time history of the pitch and heave motions in regular waves.

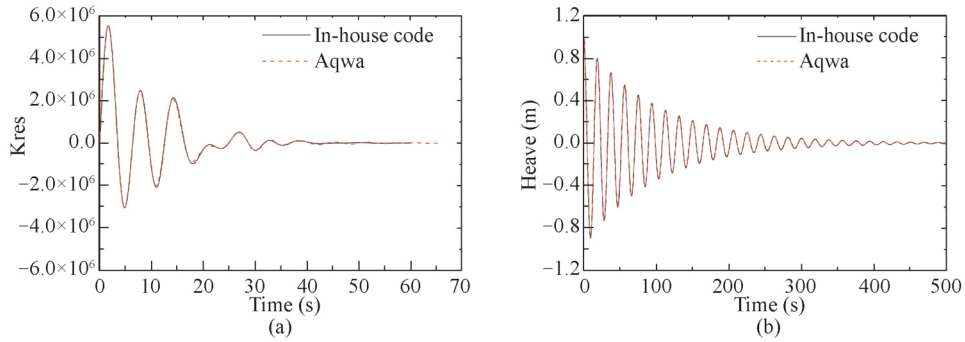


Fig. 7. Kernel function of the convolution (a) and time history of the heave motion in still water (b).

$$A = 1 - 0.287 \ln \gamma.$$

(8)

The JONSWAP spectrum's wave energy typically concentrates within a relatively narrow frequency band, which determines the characteristic wave pattern. The Equal-Energy method is implemented in the in-house code, with each $\Delta\omega_j$ containing equivalent wave energy. Consequently, the wave amplitude can be expressed as:

$$a_j = \sqrt{2S(\omega_j)\Delta\omega_j}. \quad (9)$$

A comparison of results between the in-house code and two distinct versions of Aqwa, with surge, sway and yaw motions constrained, is presented in Fig. 8. The simulation parameters include a peak enhancement factor of 3.3, peak frequency of 0.2 rad/s, and significant wave height of 3.0 m. Table 2 presents the statistical values of the heave and pitch motions. The numerical results for irregular wave cases from the in-house code demonstrate strong agreement with the commercial software, validating the reliability of the in-house code.

4 Results and discussions

4.1 Numerical and experimental results with impact force

This study conducted two launch tests in the seakeeping basin: one in still water and another in regular waves. The numerical results from the in-house code simulating the launch process were compared with experimental data to validate the numerical method and examine the hydrodynamic characteristics of the launch platform under significant impact force. Impact forces were measured using a force sensor mounted between the rocket and platform. The force sensor was positioned at coordinates (28.7 m, 0 m, 30 m), designated as the launch point in this study.

Fig. 9 illustrates the impact force measured during the first launch test. The measured impact force, represented by the red line, persists for 1.9 s. With the launch start time defined as $t_{\text{case1-1}} = 127.2$ s, the platform experiences maximum force at $t_{\text{case1-2}} = t_{\text{case1-1}} + 0.6$ s, and the launch concludes at $t_{\text{case1-2}} = t_{\text{case1-1}} + 1.9$ s. The measured maximum force reaches 2.2×10^7 N, while the rocket weight is 2.36×10^6 N.

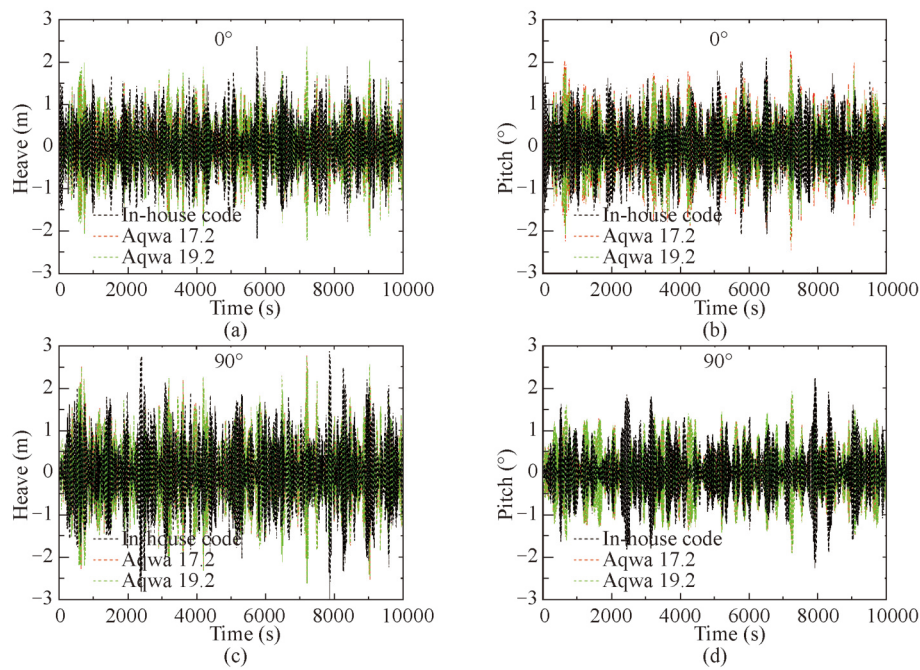


Fig. 8. Time history of the pitch and heave motions in irregular waves.

Table 2 Statistics values in irregular waves

Aqwa vs. in-house code	Wave direction (°)	Aqwa17.2	Aqwa19.2	In-house code
Significant values of heave motions (m)	0	1.15608	1.23312	1.20524
Significant values of pitch motions (°)	0	1.24421	1.14623	1.19028
Significant values of heave motions (m)	90	1.46809	1.46529	1.49690
Significant values of pitch motions (°)	90	1.07437	1.07672	1.05962

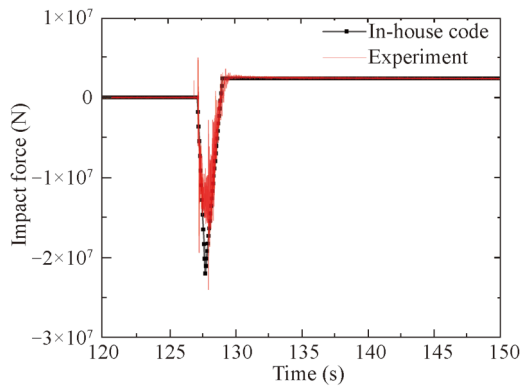


Fig. 9. Impact force measured during the first test and utilized in numerical simulation.

The sampling frequencies of impact force and motion are 15000 and 8 Hz, respectively. The black line represents the simplified impact force used in numerical simulations to correspond with the measured impact force. Platform motions were recorded using the 6-DOF non-contact motion measurement system. Given the dynamic positioning system implementation and the rocket's central line positioning, the horizontal plane and roll motion responses were minimal. Therefore, the analysis focuses on the platform's heave and pitch motions under launch impact force. Fig. 10 presents

the time history of the heave and pitch motions in still water with impact force. The experimental data demonstrates strong correlation with the numerical results. Post-impact motions calculated by the in-house code and Aqwa are identical, with artificial diagonal linear damping values of 2.00×10^6 N/(m/s) and 0.7×10^{10} Nm/(rad/s) for heave and pitch motions, respectively, based on experimental data and Eq. (14.17) in Aqwa theory manual (Ansys, 2024). Fig. 10 additionally reveals that post-launch heave and pitch motion periods are 17.5 and 17.6 s respectively, substantially exceeding the launch impact force duration.

The impact force measured in the second test under regular wave conditions is presented in Fig. 11. The wave parameters include a period of $T = 6$ s, wave height of 2.5 m, and wave direction of 135° . The impact force persisted for 1.9 s, with the initial time designated as $t_{\text{case2-1}}$. The maximum measured force reached 2.0×10^7 N at $t = t_{\text{case2-1}} + 0.6$ s, while the rocket weight was maintained at 2.45×10^6 N from $t = t_{\text{case2-1}} + 1.9$ s. Fig. 12 illustrates the time history of the launch platform's heave and pitch motions under regular wave conditions. The experimental data are represented by blue lines, while black and red lines indicate numerical results with and without the impact force, respectively. The numerical results demonstrate strong correlation with the experimental data initially; however, notable discrepancies

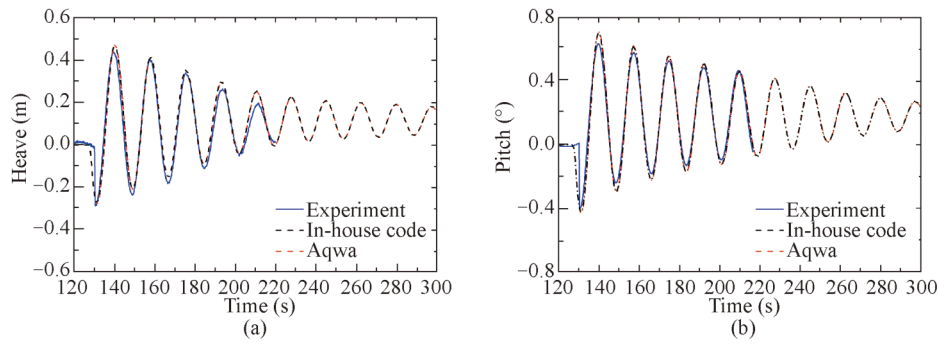


Fig. 10. Time history of the heave and pitch motions in still water with the impact force.

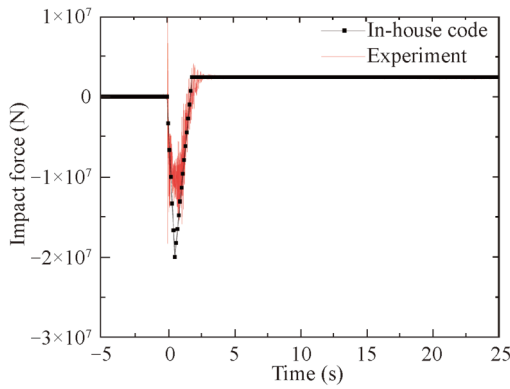


Fig. 11. Impact force measured in the second test and utilized in numerical analysis.

emerge after approximately 6–7 motion periods, likely attributable to wave reflections from the wave basin boundaries reaching the platform.

For the launch case, the platform motion was calculated in time domain based on Eq. (4), accounting for both the wave excitation force and the non-periodic impact force presented in Fig. 9 and Fig. 11. Fig. 13 reveals an notable observation: the platform motions in waves under substantial impact force can be separated into two components: the motion in regular waves without impact force, and the motion in still water with impact force. For the specifically designed launch platform and sea conditions presented in this paper, the motions induced by the impact force and

rocket weight significantly exceed the wave-induced response in both heave and pitch directions.

4.2 Hydrodynamic characteristic in moderate sea state

The launch platform is engineered to execute launch missions in low to moderate sea states, necessitating an examination of its hydrodynamic characteristics and operational performance under these conditions.

The significant wave heights in moderate sea states are assumed to be within the range of $1.25 \text{ m} < H_s < 2.5 \text{ m}$, the corresponding wave spectral peak periods are determined by the commonly used empirical formula below

$$\sqrt{13H_s} < T_p < \sqrt{30H_s}. \quad (10)$$

To eliminate uncertainties in the statistical results of irregular wave simulations, the aforementioned sea states for irregular waves were transformed into regular waves with equivalent severity. This transformation was accomplished by establishing regular wave height and wave periods of $H = \sqrt{2}H_{s\max} \approx 3.5 \text{ m}$ as standard derivation of the trigonometric function $\sigma = \sqrt{2}/2$ and $T=8.7, 7.2$, and 5.7 s to encompass the specified wave period range. The simulations considered platform responses in various wave directions, including cases without launch and with launch at different temporal points relative to platform motion, as detailed in Table 3.

The temporal comparison of the heave and pitch motions during launches at varying moments is illustrated in Fig. 14. As previously analyzed, the launch platform's

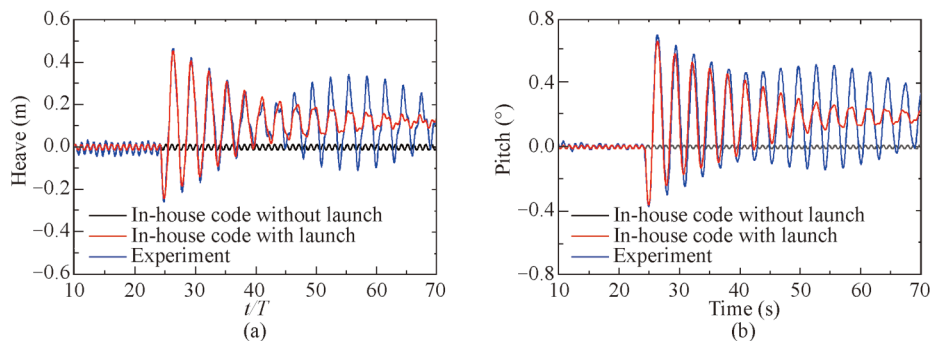


Fig. 12. Time history of the heave and pitch motions in regular waves, comparing cases with and without impact force.

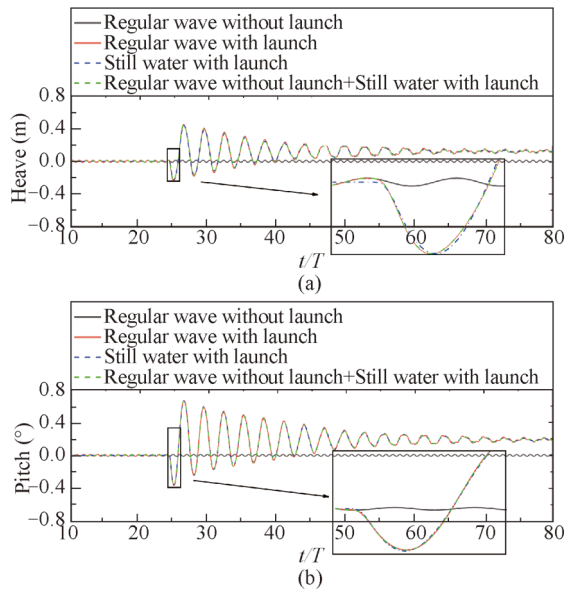


Fig. 13. Time history of the heave and pitch motions in still water and in regular wave with and without the impact force.

motions during the launch operation comprise two components: motions induced solely by waves and those caused by the combined impact force and rocket weight. Fig. 14 demonstrates that the motions during launch represent superpositions of decay and sinusoidal curves. The maximum heave motion occurs when the peak of wave-induced motion aligns with the peak of impact force-induced heave motion, as observed at $T=8.7$ s, where the maximum wave-

Table 3 Simulation cases in regular waves in Eq. (6)

Cases-Re	H (m)	T (s)	Wave direction ($^{\circ}$)	Launch	Launch moment
1	3.5	5.7	0	No	—
2	3.5	7.2	0	No	—
3	3.5	8.7	0	No	—
4	3.5	5.7	0	Yes	Heave peak
5	3.5	5.7	0	Yes	Heave zero
6	3.5	5.7	0	Yes	Heave trough
7	3.5	7.2	0	Yes	Heave peak
8	3.5	7.2	0	Yes	Heave zero
9	3.5	7.2	0	Yes	Heave trough
10	3.5	8.7	0	Yes	Heave peak
11	3.5	8.7	0	Yes	Heave zero
12	3.5	8.7	0	Yes	Heave trough
13	3.5	8.7	45	Yes	Heave trough
14	3.5	8.7	90	Yes	Heave trough
15	3.5	8.7	135	Yes	Heave trough
16	3.5	8.7	180	Yes	Heave trough
17	4.5	8.7	0	No	—
18	4.5	8.7	0	Yes	Heave trough

induced heave motion measures 0.07 m while the total maximum heave motion reaches 0.53 m. Similarly, regarding pitch motions, the maximum peak value of the wave-induced component measures 0.2° while the total maximum value during the launching process remains below 1° . These findings demonstrate that this self-designed platform exhibits robust sea-keeping performance in medium sea states and proves suitable for conducting sea-launch operations.

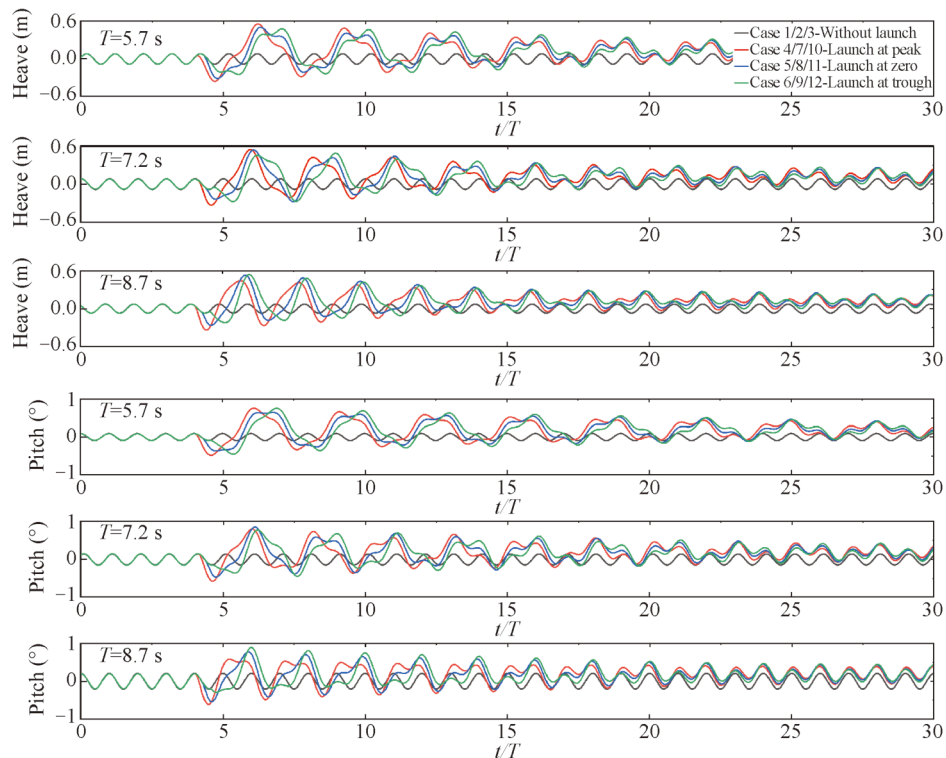


Fig. 14. Time history of the heave and pitch motions during launch at different moments.

Given that the heave and pitch motions exhibit greater magnitudes at $T = 8.7$ s, additional simulations were conducted to analyze the motions across various wave directions at this specific period. The analysis reveals no substantial variations in the launch process across different wave directions, as illustrated in Fig. 15. The heave and pitch motions remain constrained below 0.6 m and 1° , respectively. The roll motions maintain consistency throughout the launching processes, as the launch impact force acts along the central line of the platform and does not influence the platform's roll response due to its symmetrical configuration about the X axis.

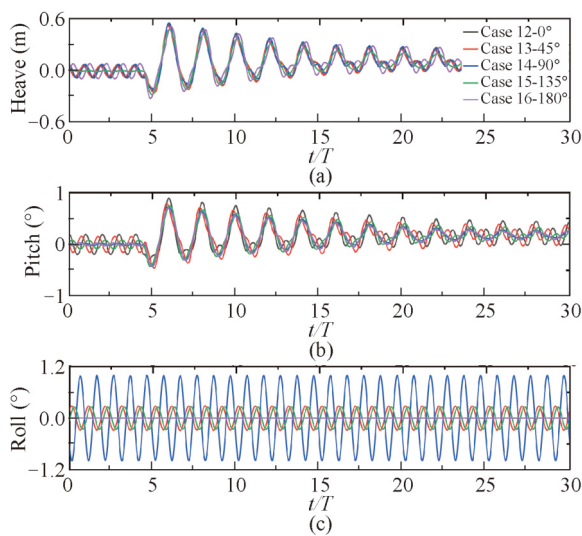


Fig. 15. Time history of the heave, pitch and roll motions during launch in different wave directions.

The time history of heave and pitch motions during the launch process at varying wave heights is illustrated in Fig. 16. The analysis reveals that the impact force-induced component predominates, while wave-induced responses in

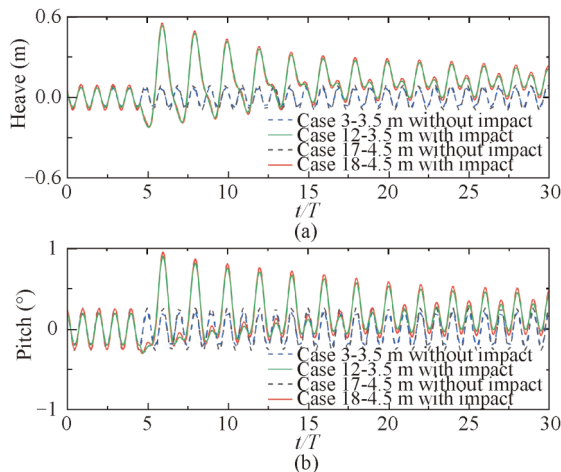


Fig. 16. Time history of the heave and pitch motions during launch at different wave heights.

heave and pitch demonstrate minimal variation between $H=3.5$ and 4.5 m.

Simulations for irregular wave cases were conducted, with wave parameters presented in Table 4. The motion responses are illustrated in Fig. 17. Consistent with the regular wave cases, the impact force-induced motions remain predominant, while the maximum heave and pitch motions maintain relatively small amplitudes, remaining within acceptable parameters for sea launch operations.

Table 4 Simulation cases in irregular waves

Case-Ir	H_s (m)	T_p (s)	Wave direction ($^\circ$)	Launch
19	2.5	5.7	0	No
20	2.5	5.7	0	Yes
21	2.5	5.7	45	No
22	2.5	5.7	45	Yes
23	2.5	5.7	90	No
24	2.5	5.7	90	Yes
25	2.5	5.7	135	No
26	2.5	5.7	135	Yes
27	2.5	5.7	180	No
28	2.5	5.7	180	Yes

The simulation results demonstrate that the self-designed platform exhibits favorable response characteristics in moderate sea states, establishing its suitability as a facility for conducting sea launch missions.

5 Conclusions

This study presents the development of an in-house code based on linear three-dimensional potential theory to simulate platform motion responses under transient impact force during sea-launch operations. The implementation of convolution integral for both regular and irregular wave simulations represents an innovative approach compared to existing software, particularly in handling time-varying non-periodic impact forces. Two launch model experiments were conducted in still water and regular waves. The numerical results from the in-house code demonstrate good agreements with the model experimental data. Analysis reveals that platform motions comprise wave-induced and impact force-induced components, with the latter being predominant during the launching process. Further investigation of the self-designed sea launch platform in moderate sea states using the in-house code numerically demonstrates that sea launch missions can be conducted safely, with maximum heave motion remaining below 0.6 m and maximum pitch motion not exceeding 1° . It should be noted that this study assumes very high structural stiffness, and structural deformations are not considered in the analysis.

Authors' contribution

XIE Chun-mei: Conceptualization, Methodology, Validation, Formal analysis, Data curation, Writing – original draft preparation, Writing – review and editing. YANG Jun-cheng: Investigation, Formal analysis, Writing – original

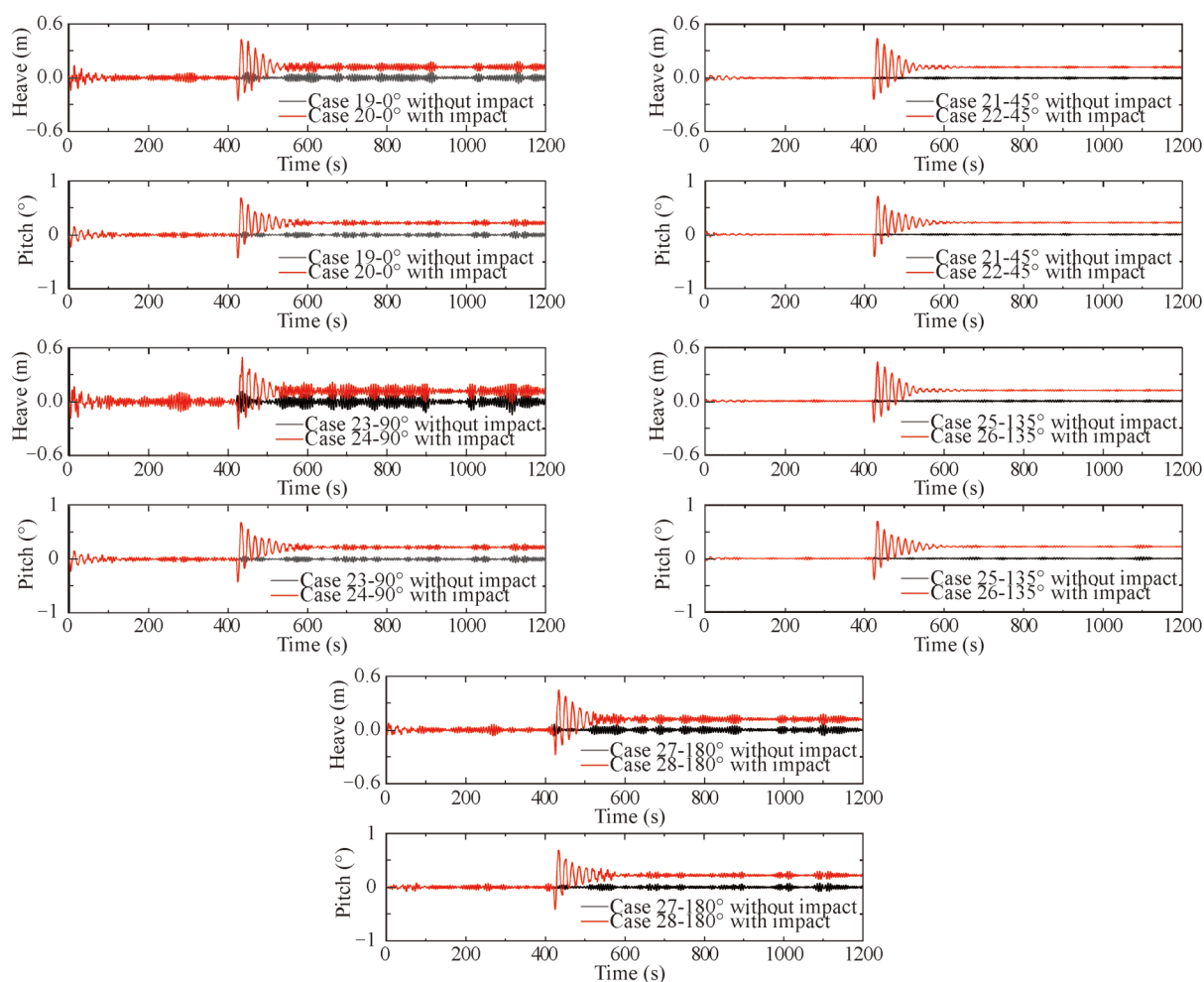


Fig. 17. Time history of the heave and pitch motions during launch in irregular waves under different wave direction angles.

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Data availability

Data sets generated during the current study are available from the corresponding author on reasonable request.

Competing interests

The authors declare no competing interests.

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